

Trainer Debrief — Lab 27

Statistical Process Control and Chart Selection — In Depth

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Select and construct the correct control chart for your capstone Y (A5, K1).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-27-spc-control-charts.xlsx

XbarR: chart the subgroup means. Centre line ~221 kPa, UCL ~233.5, LCL ~208.5. Subgroups 24 and 25 sit ABOVE the UCL — a genuine special cause (a fixture reset during that shift). Investigate and exclude before computing capability. With those points removed the process is stable: Ppk ~ 1.40, up from the 0.62 baseline in Lab 14. I-MR: clamp pressure is in control at ~47.4 bar (the Lab 25 optimum), so the control plan is holding the X. Use I-MR, not Xbar-R — these are individual readings with no rational subgroup. pChart: an obvious sustained step down at week 13 when the improvement went live — 2.9% to 1.2% defective. Recalculate the control limits AFTER week 13; carrying the old limits forward is a classic SPC error.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your chart selection is justified against the selection logic, limits are computed from 25+ subgroups with the correct constants, and all run rules are applied with every signal investigated.

Where this sits in the capstone

A full re-derivation of SPC — chart anatomy, the selection logic, control limits and the out-of-control rules. You must be able to teach this to Green Belts and audit their chart choices, which is a core Black Belt accountability. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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